UNIVERSITÄT
BERN

Applied Optimization
Solution to Exercise 1 - Convex Sets

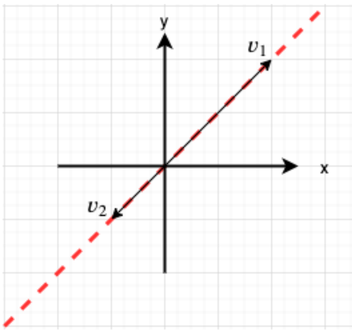
Convex Sets (1 pt)

Example sets

Sketch the following sets in $\mathbb{R}^2$

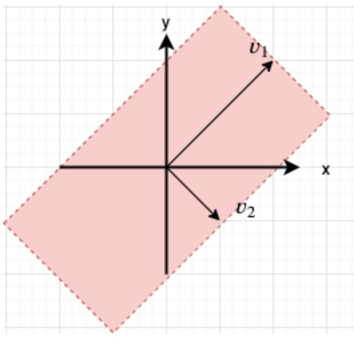
1. $\operatorname{span}\left\{\begin{pmatrix}1\\1\end{pmatrix},\begin{pmatrix}-0.5\\-0.5\end{pmatrix}\right\}$

   Solution: $\operatorname{span}\{v_1,v_2\} = \alpha_1 v_1 + \alpha_2 v_2$ with $\alpha_1,\alpha_2 \in \mathbb{R}$. Thus the set is line along colinear vectors v_1, v_2.



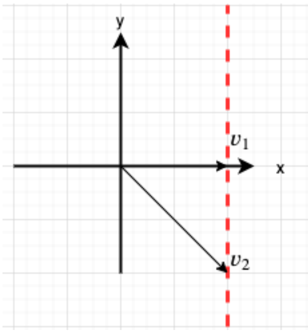
2. $\operatorname{span}\left\{\begin{pmatrix}1\\1\end{pmatrix},\begin{pmatrix}0.5\\-0.5\end{pmatrix}\right\}$

   Solution: $\operatorname{span}\{v_1,v_2\} = \alpha_1 v_1 + \alpha_2 v_2$ with $\alpha_1,\alpha_2 \in \mathbb{R}$ as before, but now v_1 and v_2 are not colinear anymore, thus they span the full $\mathbb{R}^2$ plane.



3. $\operatorname{aff}\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}1\\-1\end{pmatrix}\right\}$

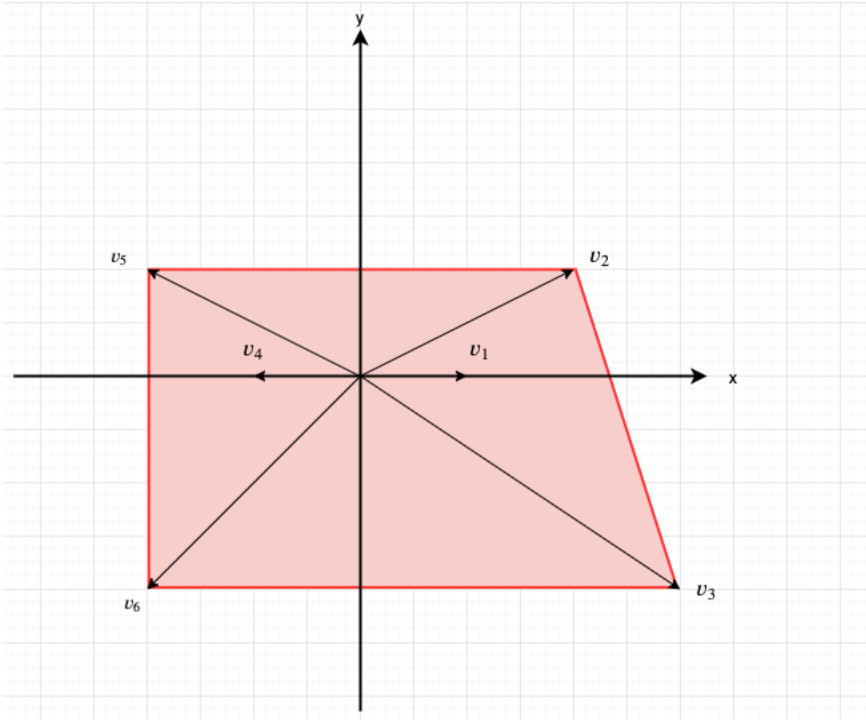
   Solution: $\operatorname{aff}\{v_1,v_2\} = \alpha_1 v_1 + \alpha_2 v_2$ with $\alpha_1,\alpha_2 \in \mathbb{R}$ as before. Subject to the additional constraint $\alpha_1 + \alpha_2 = 1$. Thus the set is the line passing through the tips of the vectors v_1, v_2.



4. $\operatorname{conv}\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}2\\1\end{pmatrix},\begin{pmatrix}3\\-2\end{pmatrix},\begin{pmatrix}-1\\0\end{pmatrix},\begin{pmatrix}-2\\1\end{pmatrix},\begin{pmatrix}-2\\-2\end{pmatrix}\right\}$

   Solution:

   By the definition of convex combination $w = \sum_{i=1}^{6} \alpha_i v_i$ with $\alpha_i \in R$, $\sum_{i=1}^{6} \alpha_i = 1$ and additionally $\alpha_i \geq 0, i = 1 \ldots 6$. Thus the set is the "convex hull of the vector tips".



Convexity

Let $C \in \mathbb{R}^n$ be a convex set, with $x_1, \ldots, x_k \in C$, and let $\theta_1, \ldots, \theta_k \in \mathbb{R}$ satisfy $\theta_i \geq 0, \theta_1 + \ldots + \theta_k = 1$. Show that $\theta_1 x_1 + \ldots + \theta_k x_k \in C$. (The definition of convexity is that this holds for $k = 2$; you must show it for arbitrary k.) Hint. Use induction on k.

Solution:

We illustrate the case $k = 3$ and then show the general case for $k > 3$ by induction.

First suppose $\theta_1, \theta_2, \theta_3 \geq 0$. To show that $y = \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 \in C$ first note that at least one of the θ_i is different from 1. Without lost of generality assume $\theta_1 \neq 1$, thus we can write:

$$y = \theta_1 x_1 + (1 - \theta_1)(\mu_2 x_2 + \mu_3 x_3)$$

with $\mu_2 = \frac{\theta_2}{1-\theta_1}$ and $\mu_3 = \frac{\theta_3}{1-\theta_1}$.

Note that $\mu_2, \mu_3 \geq 0$ and

$$\mu_2 + \mu_3 = \frac{\theta_2}{1-\theta_1} + \frac{\theta_3}{1-\theta_1} = \frac{\theta_2 + \theta_3}{1-\theta_1} = \frac{1-\theta_1}{1-\theta_1} = 1$$

Since C is convex and $x_2, x_3 \in C$, we conclude that $\mu_2 x_2 + \mu_3 x_3$ is in C because it is a convex combination of elements in C.

Since $\mu_2 x_2 + \mu_3 x_3$ is in C and x_1 is also in C, the y is also in C because it is a convex combination of elements in C.

The induction for $k > 3$ can be performed similarly. Assume $x_1, \ldots x_{k-1} \in C$, and its convex combination $y_{k-1} = \mu_1 x_1 + \ldots \mu_{k-1} x_{k-1}$ with $\mu_1 + \ldots \mu_{k-1} = 1$, and $\mu_i \geq 0, i = 1, \ldots k-1$ is also in C.

For the induction step pick an $x_k \in C$ with $\theta_k \neq 1$, $\theta_k \geq 0$. Thus since y_{k-1} is in C we can write

$$y_k = \theta_k x_k + (1 - \theta_k) y_{k-1} = \theta_k x_k + (1 - \theta_k)(\mu_1 x_1 + \cdots + \mu_{k-1} x_{k-1}) = \sum_{i=1}^{k} \theta_i x_i$$

with $\theta_i = \mu_i (1 - \theta_k)$ for $i = 1 \ldots k-1$. Note that

$$\sum_{i=1}^{k} \theta_i = \theta_k + (1 - \theta_k) \sum_{i=1}^{k-1} \mu_i = \theta_k + (1 - \theta_k) \cdot 1 = 1$$

Thus we and conclude that y_k is in C because it is a convex combination of elements in C.

Linear Equations

Show that the solution set of linear equations $\{x|Ax = b\}$ with $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ is an affine set.

Solution:

Alternative to the proof in the lecture slides, on can prove that a linear affine combination of two elements of the solution set is also in the solution set.

Let $x_1, x_2 \in \{x|Ax = b\}$, and $\alpha \in \mathbb{R}$

$\Rightarrow (1-\alpha)Ax_1 + \alpha Ax_2 = (1-\alpha)b + \alpha b = b.$

$\Rightarrow$ the affine combination $(1-\alpha)x_1 + \alpha x_2$ is also a solution.

Linear Inequations

1. Show that the solution set of linear inequations $\{x|Ax \preceq b, Cx = d\}$ with $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k\times n}$ and $d \in \mathbb{R}^k$ is a convex set. Here $\preceq$ means componentwise less or equal.

   Solution:

   To show convexity we can check whether the convex combination of two elements of the solution set is also in the solution set.

   Let $x_1, x_2 \in \{x|Ax \preceq b, Cx = d\}$

   $\Rightarrow Ax_1 \preceq b \wedge Cx_1 = d \wedge Ax_2 \preceq b \wedge Cx_2 = d$

   Define $w = (1-\alpha)x_1 + \alpha x_2,\ \alpha \in [0,1]$

   $\Rightarrow$

   $$\begin{aligned} Aw &= A(1-\alpha)x_1 + A\alpha x_2 \\ &= \underbrace{(1-\alpha)}_{\in[0,1]}\underbrace{Ax_1}_{\preceq b} + \alpha\underbrace{Ax_2}_{\preceq b} \preceq (1-\alpha)b + \alpha b = b \end{aligned}$$

   Similarly

   $$Cw = (1-\alpha)Cx_1 + \alpha Cx_2 = (1-\alpha)d + \alpha d = d$$

2. Is it an affine set?

Solution:

To check if the set is affine, we attempt to make an affine combination of two elements in the solution set and check if it belogs to the solution set.

Define $v = (1-\alpha)x_1 + \alpha x_2$, $\alpha \in \mathbb{R}$ (unconstrained).

$Cv = C(1-\alpha)x_1 + C\alpha x_2 = (1-\alpha)d + \alpha d = d$ is fine.

but

$Av = \underbrace{(1-\alpha)}_{\in\mathbb{R}}\underbrace{Ax_1}_{\preceq b} + \alpha\underbrace{Ax_2}_{\preceq b}$

Since α is unconstrained, it does not hold that $Av \preceq b$ anymore. The solution set is not affine.

Voronoi description of halfspace

Let a and b be distinct points in $\mathbb{R}^n$. Show that the set of all points that are closer (in Euclidean norm) to a than b, i.e., $\{x | \|x-a\|^2 \leq \|x-b\|^2\}$, is a halfspace. Describe it explicitly as an inequality of the form $c^T x \leq d$. Draw a picture.

Solution:

It suffices to multiply out the norm square terms and simplify to transform $\|x-a\|^2 \leq \|x-b\|^2$ in $Cx \leq d$.

$$
\begin{aligned}
(x-a)^T(x-a) &\leq (x-b)^T(x-b) \\
x^T x - 2a^T x + a^T a &\leq x^T x - 2b^T x + b^T b \\
\underbrace{2(b^T - a^T)}_{C} x &\leq \underbrace{b^T b - a^T a}_{d}
\end{aligned}
$$

In the above simplification the symmetry of scalar products was used. i.e. $a^T x = x^T a$

Convex Illumination Problem (1 pt)

Show that the solution $p^* = (p_1^*, p_2^*, \ldots, p_n^*)^T \in \mathbb{R}^n$ of the non-convex illumination problem from the lecture

$$
\begin{array}{ll}
\text{minimize} & \max\limits_{k=1\ldots m} |\log I_k - log I_{des}| \\
\text{subject to} & 0 \leq p_j \leq p_{max}, \quad j = 1 \ldots n
\end{array}
$$

with $I_k = \sum_{j=1}^n a_{kj} p_j$ for geometric constants $a_{jk} \in \mathbb{R}$, a constant desired illumination $I_{des} \in \mathbb{R}$ and an upper bound $p_{max} \in \mathbb{R}$ on the lamp power, is identical to the solution of the following equivalent (convex) problem

$$
\begin{array}{ll}
\text{minimize} & \max\limits_{k=1\ldots m} h(I_k / I_{des}) \\
\text{subject to} & 0 \leq p_j \leq p_{max}, \quad j = 1 \ldots n
\end{array}
$$

with $h(u) = \max\{u, 1/u\}$.

Solution:

Let p^* be the optimal or solution of the problem. The transformations applied to the cost function may change the values but not the solution p^*. i.e. the position of the minimizer.

$$
\begin{aligned}
p^* &= \arg\min_p \quad \max_{k=1\ldots m} |\log I_k - \log I_{des}| \\
&= \arg\min_p \quad \max_{k=1\ldots m} |\log(I_k/I_{des})| && \text{property of the logarithm} \\
&= \arg\min_p \quad \exp\left(\max_{k=1\ldots m} |\log(I_k/I_{des})|\right) && e^x \text{ is strictly increasing} \\
&= \arg\min_p \quad \max_{k=1\ldots m} \exp(|\log u|) && \text{also because } e^x \text{ is strictly increasing. } u = I_k/I_{des}
\end{aligned}
$$

let

$$
\begin{aligned}
h(u) &= \exp(|\log u|) \\
&= \begin{cases} \exp(\log u) = u & \text{for } u \geq 1 \\ \exp(-\log u) = \exp(\log(1/u)) = \frac{1}{u} & \text{for } u < 1 \end{cases}
\end{aligned}
$$

$$
\Rightarrow \quad p^* = \arg\min_p \max_{k=1\ldots m} h(I_k/I_{des})
$$

with

$$
h(u) = \begin{cases} u & u \geq 1 \\ \frac{1}{u} & u < 1 \end{cases}
$$

which is equivalent to $h(u) = \max(u, 1/u)$.